

离线RAG

```
In [2]: from langchain_community.document_loaders import TextLoader
        from langchain_text_splitters import RecursiveCharacterTextSplitter
        from langchain_huggingface import HuggingFaceEmbeddings
        from langchain_chroma import Chroma
```

C:\Users\yanchen\AppData\Local\Temp\ipykernel_31600\3182895319.py:1: DeprecationWarning: `langchain-community` is being sunset and is no longer actively maintained. See <https://github.com/langchain-ai/langchain-community/issues/674> for details and migration guidance toward standalone integration packages.

```
from langchain_community.document_loaders import TextLoader
```

读取外部知识

```
In [6]: loader = TextLoader(
        ". /普通高等学校学生管理规定.txt",
        encoding="utf-8"
    )

documents = loader.load()
```

进行分块Chunks

```
In [7]: text_splitter = RecursiveCharacterTextSplitter(
        chunk_size=500, # 每个文本块的大小
        chunk_overlap=50, # 文本块之间的重叠部分
    )
doc_splits = text_splitter.split_documents(documents)
```

将 chunks 进行 BM25 编码

```
In [14]: import jieba
        from rank_bm25 import BM25Okapi

pdf_content_words = [jieba.lcut(doc.page_content) for doc in doc_splits]#jieba是用牙
bm25_model = BM25Okapi(pdf_content_words)
```

Building prefix dict from the default dictionary ...
 Building prefix dict from the default dictionary ...
 Dumping model to file cache C:\Users\yanchen\AppData\Local\Temp\jieba.cache
 Dumping model to file cache C:\Users\yanchen\AppData\Local\Temp\jieba.cache
 Loading model cost 0.675 seconds.
 Loading model cost 0.675 seconds.
 Prefix dict has been built successfully.
 Prefix dict has been built successfully.

BM25 编码结果储存

```
In [16]: import os
import pickle

os.makedirs("./rag_cache", exist_ok=True)

with open("./rag_cache/bm25_model.pkl", "wb") as f:
    pickle.dump(bm25_model, f)
```

将 chunks 进行向量化

```
In [4]: # embedding_model_path = '/data/LLM/maple77/xiaobu-embedding-v2'
embedding_model_path = "BAAI/bge-small-zh-v1.5"
embeddings = HuggingFaceEmbeddings(
    model_name=embedding_model_path,
    model_kwargs={'device': 'cpu'},
    encode_kwargs={'normalize_embeddings': True}
)
```

向量化结果存入数据库

```
In [8]: vectorstore = Chroma.from_documents(
    documents=doc_splits,
    embedding=embeddings,
    persist_directory="./rag_cache/chroma_db"
)
```

```
In [9]: from huggingface_hub import snapshot_download

path = snapshot_download(
    repo_id="BAAI/bge-small-zh-v1.5",
    local_files_only=True
)

print(path)
```

C:\Users\yanchen\.cache\huggingface\hub\models--BAAI--bge-small-zh-v1.5\snapshots\7999e1d3359715c523056ef9478215996d62a620

笔记

RAG的笔记详见语雀

通过这部分代码发现自己有很多不懂的：

- 关于模型操作部分，很多参数、环境不懂